

Exam Questions

Using Calculators for Graphs, Equations & Inequalities

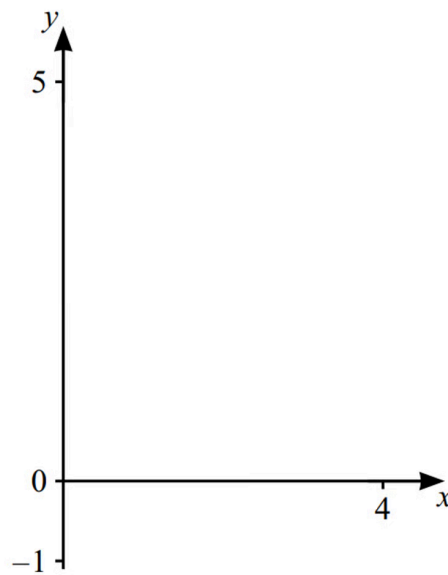
Using Calculators to Sketch Graphs / Using Calculators to Solve Equations /
Using Calculators to Solve Inequalities

Medium (5 questions)	/26
Hard (6 questions)	/47
Very Hard (4 questions)	/53
Total Marks	/126

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Medium Questions



1 (a) $f(x) = 5 - 1.25x$ $g(x) = 1.5x - 1$

On the diagram, sketch the graph of $y = f(x)$ for $0 \leq x \leq 4$.

(2 marks)

(b) On the diagram, sketch the graph of $y = g(x)$ for $0 \leq x \leq 4$.

(2 marks)

(c) Find x when $f(x) = g(x)$.

(1 mark)

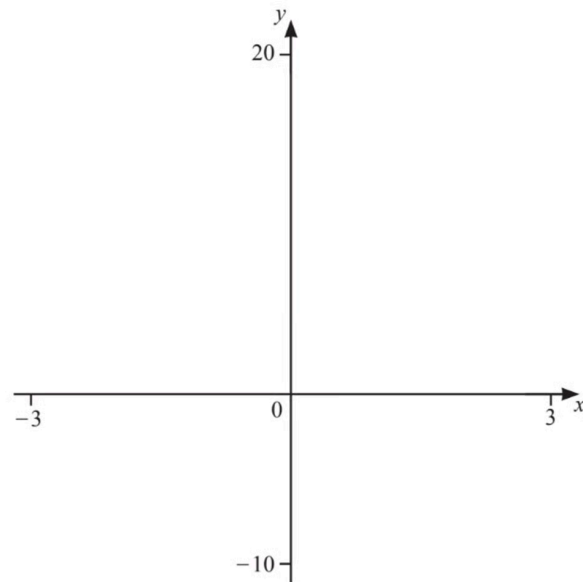
2 $2^{3x+1} = 8^{2-x}$

Find the value of x

$x = \dots\dots\dots$

(3 marks)

3 (a)



$$f(x) = x^3 - 5x + 3 \text{ for } -3 \leq x \leq 3$$

On the diagram, sketch the graph of $y = f(x)$.

(2 marks)

(b) Find the coordinates of the local maximum.

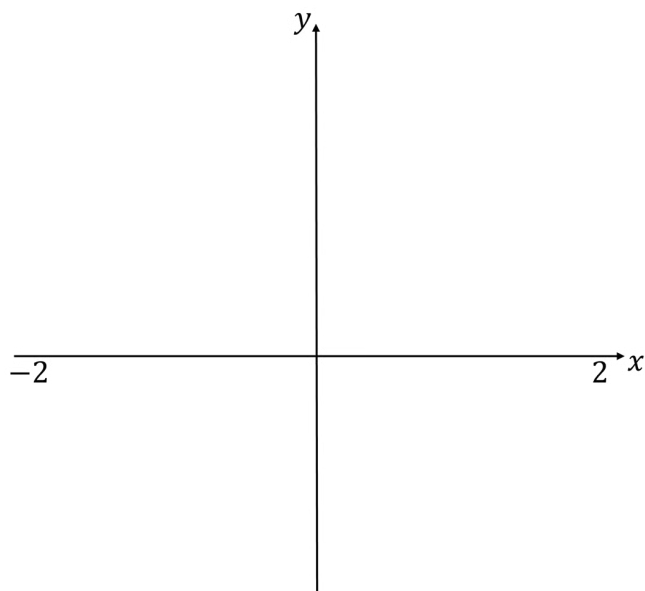
(..... ,)

(2 marks)

(c) Find the zeros of the graph of $y = f(x)$.

(3 marks)

4 (a)



$$f(x) = x^3 - 2x \quad g(x) = 1$$

On the diagram, sketch the graphs of $y = f(x)$ and $y = g(x)$ for $-2 \leq x \leq 2$.

(3 marks)

(b) Use your graphs to solve $x^3 - 2x - 1 = 0$.

(3 marks)

(c) Use your graphs to solve $x^3 - 2x > 1$.

(2 marks)

5 Use a graphical method to solve the inequality.

Show a sketch of the graph.

$$2^x + x > 5$$

(3 marks)

Hard Questions

1 (a)



$$f(x) = \frac{1}{\sin x} \text{ for } 0 \leq x \leq 360$$

On the diagram, sketch the graph of $y = f(x)$.

(3 marks)

(b) Find the coordinates of the local minimum point.

(1 mark)

(c) Write down the equations of the three asymptotes of the graph of $y = f(x)$.

(2 marks)

(d) The equation $f(x) = k$ has no solutions.

Write down the range of values of k .

(2 marks)

- (e) By sketching another graph on the diagram, solve the equation $\frac{1}{\sin x^\circ} = 5\sin\left(\frac{x}{2}\right)^\circ$ for $0 \leq x \leq 360^\circ$.

(3 marks)

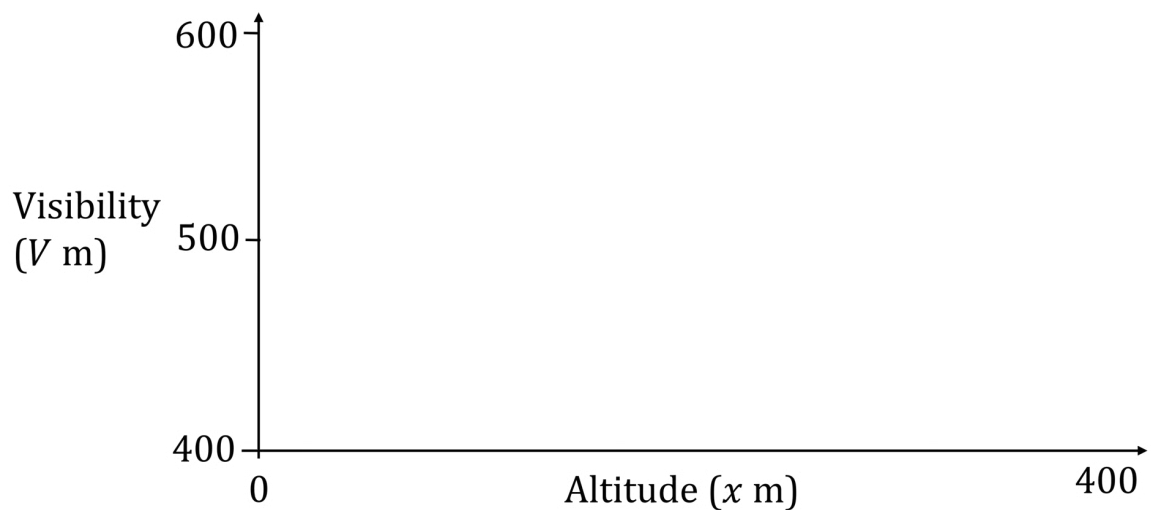
- 2 (a)** A medicine delivery company uses drones to deliver medications to remote mountain villages.

A radar system is used to control the drones, as long as the controller is within a certain distance from the drone, called the radar visibility, V metres,

A model for the radar visibility, V metres, at different altitudes, x metres up a particular mountain is

$$V = \frac{600 - x}{4} + \sqrt{70\,000 + x^2}$$

On the diagram, sketch the graph of V against x .



(4 marks)

- (b)** Find, correct to the nearest metre,

(i) the minimum visibility,

[1]

(ii) the altitude for which the visibility is at its minimum.

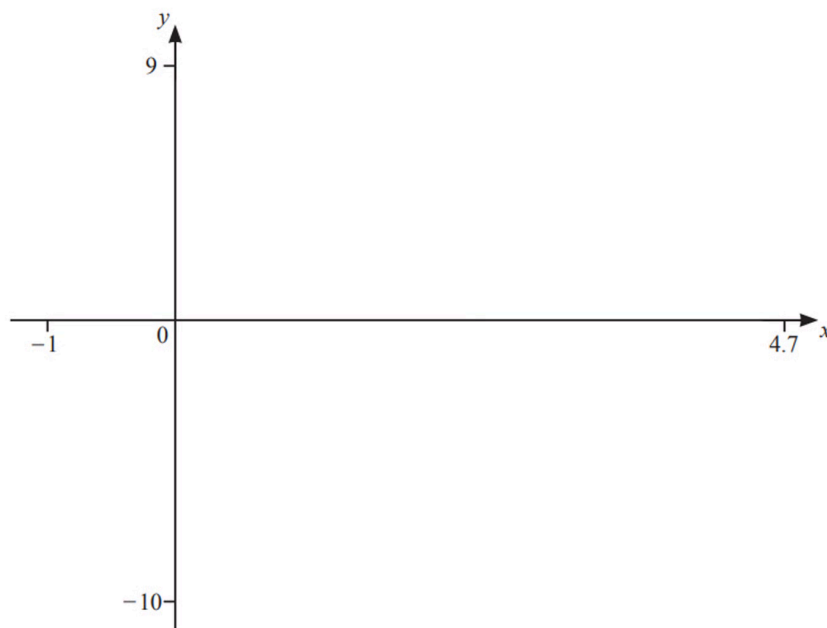
[1]

(2 marks)

- (c)** Write down a reason why the visibility given by the model might be different to the actual visibility.

(1 mark)

3 (a)



$$f(x) = (x - 2)^3 - 5x + 12 \text{ for } -1 \leq x \leq 4.7$$

On the diagram, sketch the graph of $y = f(x)$.

(2 marks)

(b) Write down the coordinates of the local maximum.

(2 marks)

(c) The equation $(x - 2)^3 - 5x + 12 = k$ has exactly 2 solutions.

Find the values of k .

(2 marks)

(d) $g(x) = -(x - 1)^2$ for $-1 \leq x \leq 4.7$

On the diagram, sketch the graph of $y = g(x)$.

(2 marks)

(e) Solve $f(x) = g(x)$.

(1 mark)

4 The curve $y = x^2 - 2x + 1$ is drawn on a grid.

A line is drawn on the same grid.

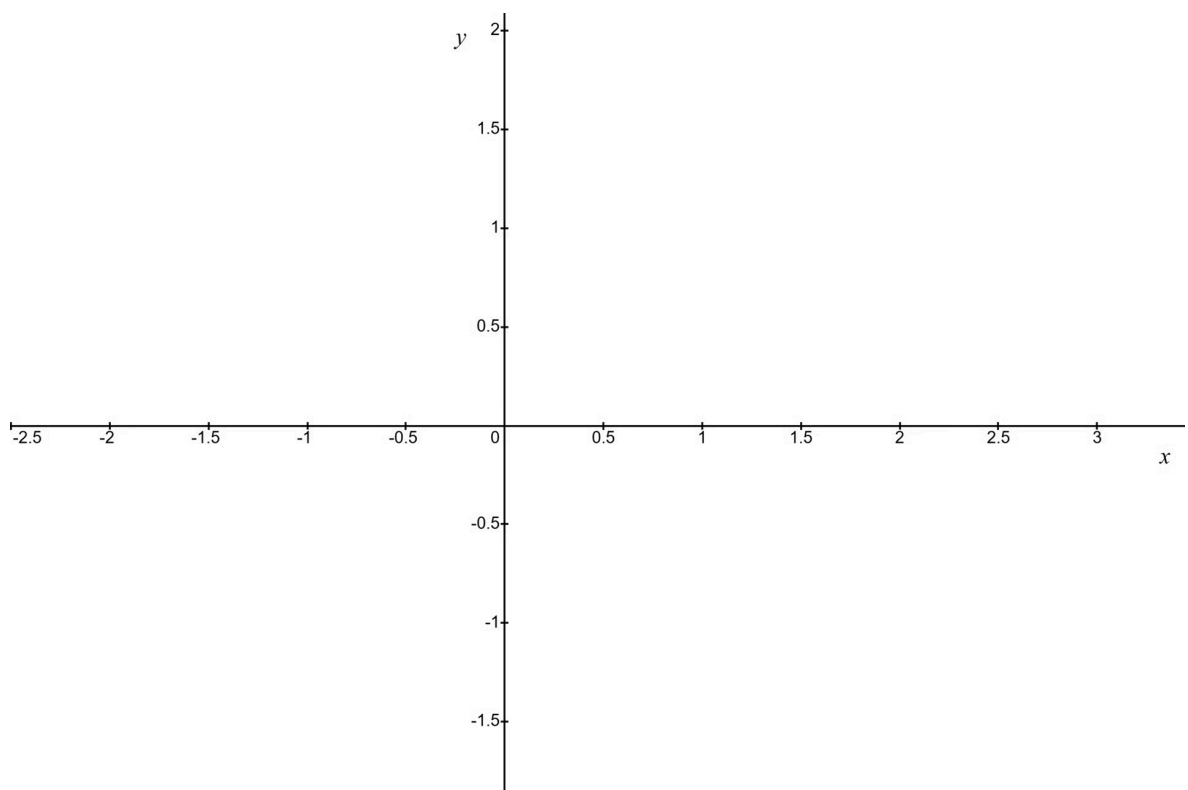
The points of intersection of the line and the curve are used to solve the equation $x^2 - 7x + 5 = 0$.

Find the equation of the line in the form $y = mx + c$.

$y = \dots\dots\dots$

(1 mark)

5 (a) On the diagram, sketch the graphs of $y = \frac{1}{x^2 + 1}$ and $y = x^3 - 2x$.



(4 marks)

(b) Solve.

$$\frac{1}{x^2 + 1} = x^3 - 2x$$

(3 marks)

(c) Solve.

$$\frac{1}{x^2 + 1} < x^3 - 2x$$

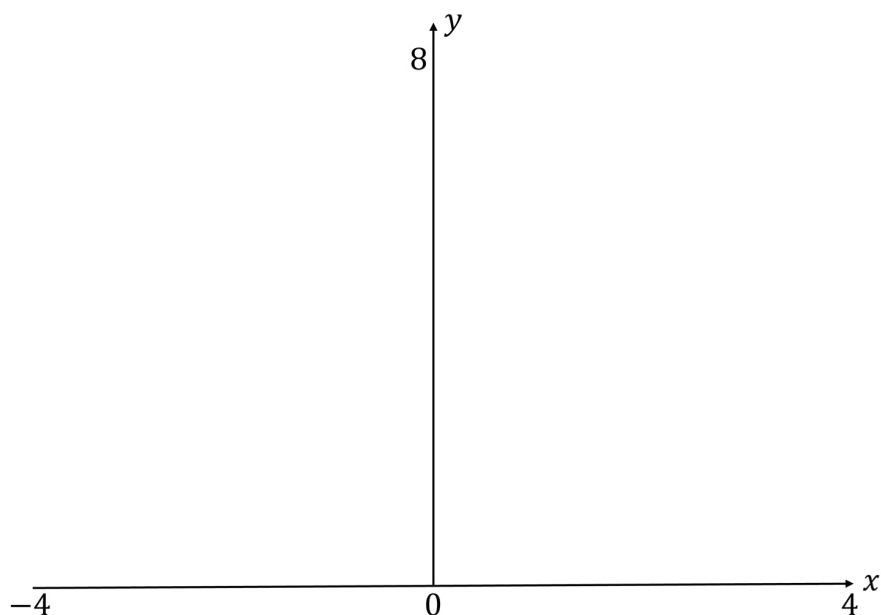
(2 marks)

(d) Find the coordinates of the minimum point of $y = x^3 - 2x$ where $x > 0$.

(1 mark)

6 (a) The graph of $y = 1 + 2^x$ has one asymptote. The equation of the asymptote is $y = 1$.

Draw the graph of $y = 1 + 2^x$ on your calculator. Use this to sketch the graph of $y = 1 + 2^x$ on the axes provided.



Label clearly the y -intercept and the asymptote.

(3 marks)

(b) Solve the following equation.

$$1 + 2^x = x + 3$$

Give your answers to 2 decimal places where appropriate.

(3 marks)

(c) Solve the following inequality.

$$1 + 2^x > x + 3$$

(1 mark)

- (d) The solutions in part (b) are also solutions to the equation below, where p and q are positive integers.

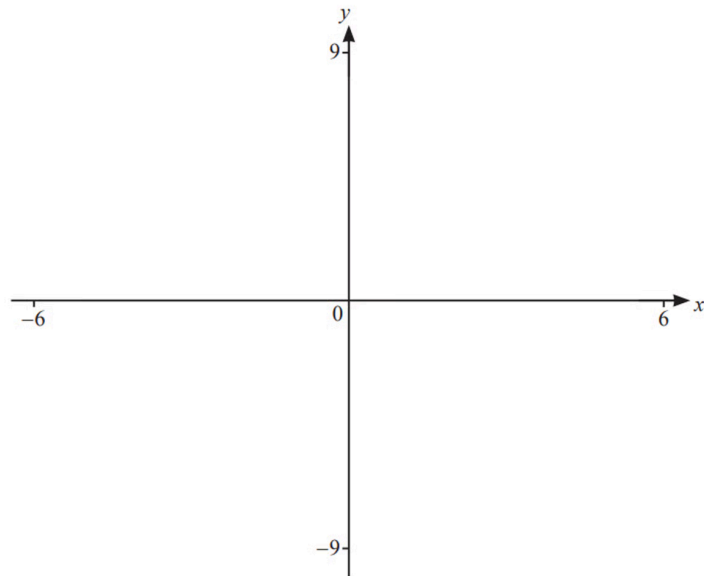
$$x = \log_p(x + q)$$

Find p and q .

(2 marks)

Very Hard Questions

1 (a)



$$f(x) = \frac{x^2 + 3x}{(x-2)(x+1)}$$

On the diagram sketch the graph of $y = f(x)$ for values of x between -6 and 6 .

(3 marks)

(b) Write down the equations of the asymptotes parallel to the y -axis.

(2 marks)

(c) Find the zeros of the graph of $y = f(x)$.

(2 marks)

(d) $g(x) = x - 3$

(i) On the diagram, sketch the graph of $y = g(x)$ for $-6 \leq x \leq 6$.

[1]

(ii) Use your graphs to solve $f(x) = g(x)$.

[3]

(iii) Solve $g(x) > f(x)$.

[3]

(7 marks)

2 (a)



$$f(x) = (\cos x^\circ)^2 \quad g(x) = 0.5 - 0.001x$$

On the diagram, sketch the graph of $y = f(x)$ for $0 \leq x \leq 360$.

(2 marks)

(b) On the same diagram, sketch the graph of $y = g(x)$ for $0 \leq x \leq 360$.

(2 marks)

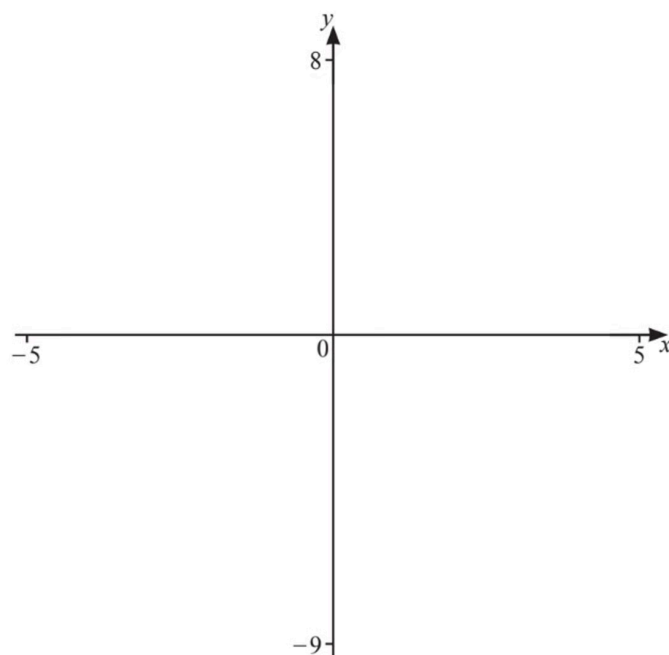
(c) Solve $f(x) = g(x)$ for $0 \leq x \leq 360$.

(4 marks)

(d) Solve $f(x) < g(x)$ for $0 \leq x \leq 360$.

(3 marks)

3 (a)



$$f(x) = x + \frac{5}{(x-2)(x+3)}$$

Sketch the graph of $y = f(x)$ for values of x between -5 and 5 .

(4 marks)

(b) Write down the equations of the asymptotes parallel to the y -axis.

(2 marks)

(c) (i) Find the coordinates of the local maximum.

(.....,) [2]

(ii) Find the coordinates of the local minimum.

(.....,) [2]

(iii) Write down the range of values of k for which $f(x) = k$ has exactly one solution.

[2]
(6 marks)

(d) $g(x) = -4 - x$

(i) Solve the equation $f(x) = g(x)$.

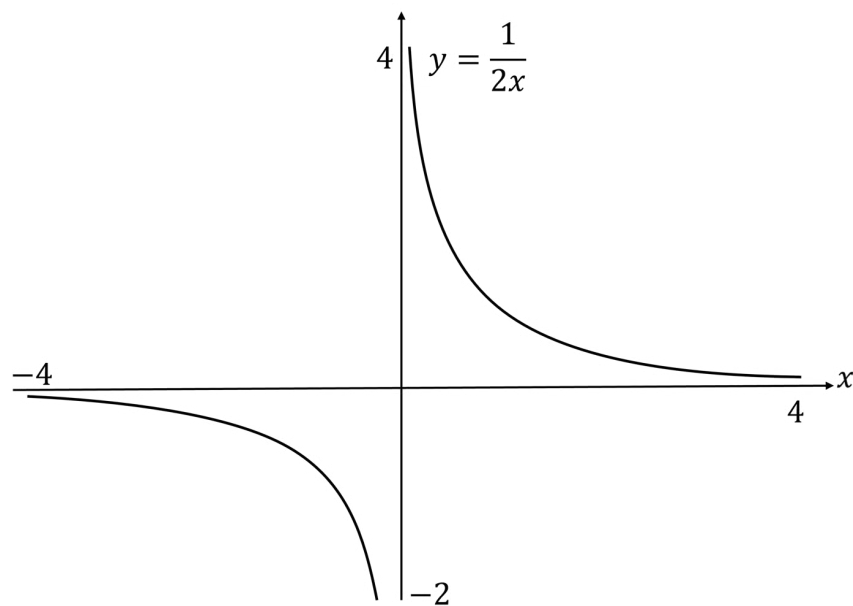
[3]

(ii) Find the solutions to the inequality $f(x) > g(x)$.

[3]

(6 marks)

- 4 (a)** The graph of $y = \frac{1}{2x}$ is shown below.



Write down the equations of any asymptotes.

(2 marks)

- (b)** On the same diagram, sketch the graph of $y = 4x^2 - x^4$.

Label clearly the coordinates of any turning points and any intercepts with the coordinate axes.

(5 marks)

- (c)** If $x > 0$, solve the following inequality.

$$4x^2 - x^4 > \frac{1}{2x}$$

Give your answer to 2 decimal places.

(3 marks)